EFU CEWS – UCAIF v3.0

Universal Country Assessment Indicator Framework

Nookratic Operational System – Kernel Documentation v1.1

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Date: March 17, 2026
Version: FINAL for Zenodo upload
DOI Parent: 10.5281/zenodo.18888082
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Keywords: EFU, CEWS, UCAIF, biophysical measurement, civilizational risk, irreversibility

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Abstract

The **EFU CEWS UCAIF v3.0** represents a comprehensive, modular framework comprising **462 quantitative indicators + 1 meta-trajectory vector** organized across 26 thematic blocks. This system employs a **"hagyma metódus" (onion methodology)** prioritizing biophysical core metrics while scaling through dynamic, entity-specific, topological, and meta-systemic layers.

Designed for universal applicability across countries, corporations, projects, communities, and crisis scenarios, UCAIF v3.0 integrates:

- **Biophysical first principles** (W_irrev irreversibility weighting 0-1000)
- **DQI v1.0 data quality indexing** (5-component transparency audit)
- **Shadow trigger detection** (greenwashing, concealed fluxes)
- **Cascade risk matrices** (systemic contagion modeling)
- **Regenerative thresholds** (700-series recovery benchmarks)

The framework emerges from 25+ years of independent development, self-financed research, and practical experience in carbon markets, crisis response, and biophysical economics.

I. NOS Kernel – System Origin

1.1 Development Context

The EFU system was created by **Simor István**, independent interdisciplinary researcher, through entirely self-financed development conducted outside institutional frameworks. This independent approach enables:

- **Biophysical first principles** over narrative compliance
- **Intention-independent measurement** (no self-reported data accepted without audit)
- **Full lifecycle flux accounting** (cradle-to-grave + latency effects)

Key practical foundations include:

- Founding and former presidency of the Hungarian Carbon Alliance
- Extensive certified training across 80+ sustainability, crisis management, and systems science courses
- Fire service operational experience informing irreversibility (W_irrev) conceptualization

1.2 Core Motivation

EFU addresses four structural failures in existing sustainability metrics:

1. **License-reality disconnect** → Biophysical primacy principle
2. **Self-reported greenwashing** → Shadow trigger matrices S01-S12
3. **Static snapshots** → Dynamic latency + recovery time layers
4. **Compliance theater** → Proof-of-burden principle + DQI v1.0

II. UCAIF v3.0 Modular Architecture

Agent Contributions Synthesis

Agent	Contribution	New Indicators
UCAIF v1.0	Biophysical core (I-VII)	210
Claude Opus	VIII-XIII modules + dynamics	+112
Agent 1	XIV-XVI: Finance/tech/food chain	+36
Agent 2	XVII-XX: System topology	+63
Agent 3	XXIII-XXV: Digital/holobiont/cultural	+24
Fire Chief	XXVI Trajectory Vector	+1 meta
TOTAL		**462 + 1 meta**

Complete 26-Block Structure

```text
BLOK A – BIOPHYSICAL CORE (I–VII) [210 indicators] 60% weight
├── I. Atmospheric, Water, Soil, Biodiversity, Climate
├── II. Antiflux (PFAS, microplastics, toxics, fossil lock-in)
├── III. Social flux (health, inequality, justice)
├── IV. Governance integrity
├── V. Economic/financial dimensions
├── VI. Geopolitical/planetary boundaries
└── VII. Meta-indicators (W_irrev, latency, cascade flags)
```

```
BLOK B – DYNAMIC LAYERS (VIII–IX) [50 indicators] 15% weight
├── VIII. Epigenetic markers
└── IX. Psycho-social dynamics + regenerative thresholds

BLOK C–D – ENTITY SPECIFIC (X–XVI) [98 indicators] 10% weight
├── X-XIII. Corporate/project/community/crisis modules
└── XIV-XVI. Financial flux + tech risks + food systems

BLOK E–F – SYSTEM TOPOLOGY (XVII–XXII) [65 indicators] 8% weight
├── XVII-XX. Energy/logistics/info/infrastructure topology
└── XXI-XXII. Network fragility + urban metabolism

BLOK G – HOLISTIC/META (XXIII–XXV) [28 indicators] 5% weight
├── XXIII-XXV. Digital flux + holobiont + cultural reproduction

XXVI. TRAJECTORY VECTOR [1 meta] 2% weight
→ COLLAPSE | STAGNANT | REGENERATIVE trajectory
```

III. Core Operational Protocols

3.1 Hagyma Metódus (Onion Prioritization)

```text
PRIORITY 1: BLOK A biophysical core → 60% weighting
PRIORITY 2: Dynamic layers + critical cascades → 25%
PRIORITY 3: Topology + entity-specific → 10%
PRIORITY 4: Meta-systemic trajectory → 5%
```

3.2 DQI v1.0 – Data Quality Index (Mandatory)

Five-component transparency audit:

```text
DQI = 0.3×Measurement + 0.25×Transparency + 0.2×Reproducibility
   + 0.15×Audit_Sovereignty + 0.1×Timeliness [0-1]
```

Concealment detection: T1(missing)/T2(withheld)/T3(destroyed) → MCI minimum enforcement

3.3 CEWS Alert Thresholds

Red Indicators %	Status	Action
<10%	□ BLUE	Monitor
10-25%	□ YELLOW	Enhanced attention
25-50%	□ ORANGE	Intervention required
>50%	□ RED	Critical
>75%	□ BLACK	Systemic collapse imminent

IV. Implementation Standards

4.1 JSON Schema Compatibility

- **205.3 schema**: Full DQI(t) timeseries + audit trail
- **900.5 Dashboard**: Real-time CEWS visualization
- **217.2.20 Source Protocol**: Tiered source hierarchy + search imperative

4.2 Fire Chief Override

Level 3 audits + T3 concealment decisions archived with Zenodo DOI stamping.

V. System Positioning

EFU CEWS UCAIF v3.0 represents an independent, biophysically-grounded measurement architecture operating outside conventional institutional frameworks. Its design priorities – biophysical reality over narrative compliance, full lifecycle accounting, and systemic irreversibility detection – emerge from 25+ years of practical development across carbon markets, crisis response, and systems science.

The framework maintains full compatibility with:
 ISO standards
 EU climate policy (taxonomy, CBAM, ETS)
 UN SDGs + Planetary Boundaries
 IPCC AR6 methodologies

Publication Status: FINAL – Ready for Nookratic Council review and international implementation partnerships.

Citation:
Simor, I. (2026). *EFU CEWS – UCAIF v3.0: Universal Country Assessment Indicator Framework*. Zenodo.

Related Works:

- Quantitative Ontology (DOI: 10.5281/zenodo.18888082)
- EFU v5.1 Diamond (parent system)
- Hungarian Carbon Alliance operational experience